

Suppose you are working in the data-driven domain at Airbnb. You have a dataset and need to derive insights from it to answer key business questions, as company officials aim to grow the business.

1. What is the distribution of listing prices?
2. How are different room types distributed?
3. How are listings distributed across different neighborhoods?
4. What is the relationship between price and room type?
5. How has the number of reviews changed over time?

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("airbnb_data.csv")
```

```
/tmp/ipykernel_8753/1490544642.py:1: DtypeWarning: Columns (25) have mixed types. Specify dtype option on import or set low_memory=False.
df = pd.read_csv("airbnb_data.csv")
```

```
df
```

	id	NAME	host id	host_identity_verified	host name	neighbourhood group	neighbourhood	lat	long	country	...	service fee	minimum nights	number of reviews
0	1001254	Clean & quiet apt home by the park	80014485718	unconfirmed	Madaline	Brooklyn	Kensington	40.64749	-73.97237	United States	...	\$193	10.0	9
1	1002102	Skylit Midtown Castle	52335172823	verified	Jenna	Manhattan	Midtown	40.75362	-73.98377	United States	...	\$28	30.0	45
2	1002403	THE VILLAGE OF HARLEM....NEW YORK !	78829239556	NaN	Elise	Manhattan	Harlem	40.80902	-73.94190	United States	...	\$124	3.0	0
3	1002755	NaN	85098326012	unconfirmed	Garry	Brooklyn	Clinton Hill	40.68514	-73.95976	United States	...	\$74	30.0	270
4	1003689	Entire Apt: Spacious Studio/Loft by central park	92037596077	verified	Lyndon	Manhattan	East Harlem	40.79851	-73.94399	United States	...	\$41	10.0	9
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
102594	6092437	Spare room in Williamsburg	12312296767	verified	Krik	Brooklyn	Williamsburg	40.70862	-73.94651	United States	...	\$169	1.0	0
102595	6092990	Best Location near Columbia U	77864383453	unconfirmed	Mifan	Manhattan	Morningside Heights	40.80460	-73.96545	United States	...	\$167	1.0	1
102596	6093542	Comfy, bright room in Brooklyn	69050334417	unconfirmed	Megan	Brooklyn	Park Slope	40.67505	-73.98045	United States	...	\$198	3.0	0
102597	6094094	Big Studio-One Stop from Midtown	11160591270	unconfirmed	Christopher	Queens	Long Island City	40.74989	-73.93777	United States	...	\$109	2.0	5
102598	6094647	585 sf Luxury Studio	68170633372	unconfirmed	Rebecca	Manhattan	Upper West Side	40.76807	-73.98342	United States	...	\$206	1.0	0

102599 rows × 26 columns

df.head()

	id	NAME	host id	host_identity_verified	host name	neighbourhood group	neighbourhood	lat	long	country	...	service fee	minimum nights	number of reviews	r
0	1001254	Clean & quiet apt home by the park	80014485718	unconfirmed	Madaline	Brooklyn	Kensington	40.64749	-73.97237	United States	...	\$193	10.0	9.0	10/1
1	1002102	Skylit Midtown Castle	52335172823	verified	Jenna	Manhattan	Midtown	40.75362	-73.98377	United States	...	\$28	30.0	45.0	5/2
2	1002403	THE VILLAGE OF HARLEM....NEW YORK !	78829239556	NaN	Elise	Manhattan	Harlem	40.80902	-73.94190	United States	...	\$124	3.0	0.0	
3	1002755	NaN	85098326012	unconfirmed	Garry	Brooklyn	Clinton Hill	40.68514	-73.95976	United States	...	\$74	30.0	270.0	7/
4	1003689	Entire Apt: Spacious Studio/Loft by central park	92037596077	verified	Lyndon	Manhattan	East Harlem	40.79851	-73.94399	United States	...	\$41	10.0	9.0	11/1

5 rows × 26 columns

```
# Columns Name - 26
df.columns
```

```
Index(['id', 'NAME', 'host id', 'host_identity_verified', 'host name',
      'neighbourhood group', 'neighbourhood', 'lat', 'long', 'country',
      'country code', 'instant_bookable', 'cancellation_policy', 'room type',
      'Construction year', 'price', 'service fee', 'minimum nights',
      'number of reviews', 'last review', 'reviews per month',
      'review rate number', 'calculated host listings count',
      'availability 365', 'house_rules', 'license'],
      dtype='object')
```

```
df.notna().sum()
```

0

id	102599
NAME	102349
host id	102599
host_identity_verified	102310
host name	102193
neighbourhood group	102570
neighbourhood	102583
lat	102591
long	102591
country	102067
country code	102468
instant_bookable	102494
cancellation_policy	102523
room type	102599
Construction year	102385
price	102352
service fee	102326
minimum nights	102190
number of reviews	102416
last review	86706
reviews per month	86720
review rate number	102273
calculated host listings count	102280
availability 365	102151
house_rules	50468
license	2

dtype: int64

Checking Missing Values

```
df.isna().sum()
```

	0
id	0
NAME	250
host id	0
host_identity_verified	289
host name	406
neighbourhood group	29
neighbourhood	16
lat	8
long	8
country	532
country code	131
instant_bookable	105
cancellation_policy	76
room type	0
Construction year	214
price	247
service fee	273
minimum nights	409
number of reviews	183
last review	15893
reviews per month	15879
review rate number	326
calculated host listings count	319
availability 365	448
house_rules	52131
license	102597

dtype: int64

```
print(df.isna().sum())
```

id	0
NAME	250
host id	0
host_identity_verified	289
host name	406
neighbourhood group	29
neighbourhood	16
lat	8

```

long                8
country             532
country code       131
instant_bookable    105
cancellation_policy 76
room type           0
Construction year   214
price              247
service fee         273
minimum nights      409
number of reviews   183
last review         15893
reviews per month    15879
review rate number   326
calculated host listings count 319
availability 365     448
house_rules         52131
license             102597
dtype: int64

```

```
df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 102599 entries, 0 to 102598
Data columns (total 26 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   id                                    102599 non-null  int64
 1   NAME                                102349 non-null  object
 2   host id                             102599 non-null  int64
 3   host_identity_verified               102310 non-null  object
 4   host name                           102193 non-null  object
 5   neighbourhood group                 102570 non-null  object
 6   neighbourhood                       102583 non-null  object
 7   lat                                 102591 non-null  float64
 8   long                                102591 non-null  float64
 9   country                             102067 non-null  object
10   country code                       102468 non-null  object
11   instant_bookable                   102494 non-null  object
12   cancellation_policy                 102523 non-null  object
13   room type                           102599 non-null  object
14   Construction year                   102385 non-null  float64
15   price                              102352 non-null  object
16   service fee                         102326 non-null  object
17   minimum nights                     102190 non-null  float64
18   number of reviews                  102416 non-null  float64
19   last review                         86706 non-null  object
20   reviews per month                   86720 non-null  float64
21   review rate number                  102273 non-null  float64
22   calculated host listings count      102280 non-null  float64
23   availability 365                    102151 non-null  float64
24   house_rules                         50468 non-null  object
25   license                             2 non-null      object
dtypes: float64(9), int64(2), object(15)
memory usage: 20.4+ MB

```

Handling Missing Values

```
df['last review'] = pd.to_datetime(df['last review'], errors = 'coerce')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 102599 entries, 0 to 102598
Data columns (total 26 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   id                                     102599 non-null  int64
1   NAME                                  102349 non-null  object
2   host id                               102599 non-null  int64
3   host_identity_verified                102310 non-null  object
4   host name                             102193 non-null  object
5   neighbourhood group                   102570 non-null  object
6   neighbourhood                         102583 non-null  object
7   lat                                   102591 non-null  float64
8   long                                  102591 non-null  float64
9   country                               102067 non-null  object
10  country code                          102468 non-null  object
11  instant_bookable                      102494 non-null  object
12  cancellation_policy                   102523 non-null  object
13  room type                             102599 non-null  object
14  Construction year                     102385 non-null  float64
15  price                                  102352 non-null  object
16  service fee                           102326 non-null  object
17  minimum nights                        102190 non-null  float64
18  number of reviews                     102416 non-null  float64
19  last review                           86706 non-null  datetime64[ns]
20  reviews per month                     86720 non-null  float64
21  review rate number                    102273 non-null  float64
22  calculated host listings count         102280 non-null  float64
23  availability 365                       102151 non-null  float64
24  house_rules                           50468 non-null  object
25  license                                2 non-null      object
dtypes: datetime64[ns](1), float64(9), int64(2), object(14)
memory usage: 20.4+ MB
```

```
df.fillna({'reviews per month':0,'last review':df['last review'].min()},inplace=True)
```

```
df.dropna(subset= ['NAME','host name'],inplace=True)
```

```
print(df.isnull().sum())
```

```
id                                     0
NAME                                   0
host id                               0
host_identity_verified                276
host name                             0
neighbourhood group                   26
neighbourhood                         16
lat                                    8
long                                   8
country                               526
country code                          122
instant_bookable                      96
cancellation_policy                   70
room type                             0
Construction year                     200
price                                  239
service fee                           268
minimum nights                        403
number of reviews                     182
```

```

last review          0
reviews per month    0
review rate number    314
calculated host listings count  318
availability 365      420
house_rules          51867
license              101947
dtype: int64

```

```

# Dropping Columns
df = df.drop(columns=['license','house_rules'],errors = 'ignore')

```

```
df.head()
```

	id	NAME	host id	host_identity_verified	host name	neighbourhood group	neighbourhood	lat	long	country	...	Construction year	price	service fee
0	1001254	Clean & quiet apt home by the park	80014485718	unconfirmed	Madaline	Brooklyn	Kensington	40.64749	-73.97237	United States	...	2020.0	\$966	\$193
1	1002102	Skylit Midtown Castle	52335172823	verified	Jenna	Manhattan	Midtown	40.75362	-73.98377	United States	...	2007.0	\$142	\$28
2	1002403	THE VILLAGE OF HARLEM....NEW YORK !	78829239556	NaN	Elise	Manhattan	Harlem	40.80902	-73.94190	United States	...	2005.0	\$620	\$124
4	1003689	Entire Apt: Spacious Studio/Loft by central park	92037596077	verified	Lyndon	Manhattan	East Harlem	40.79851	-73.94399	United States	...	2009.0	\$204	\$41
5	1004098	Large Cozy 1 BR Apartment In Midtown East	45498551794	verified	Michelle	Manhattan	Murray Hill	40.74767	-73.97500	United States	...	2013.0	\$577	\$115

5 rows x 24 columns

```
#Remove dolloar sign and convert to fload
```

```

# df['price'] = pd.to_numeric(
#     df['price'].replace('\$', '', regex=True),
#     errors='coerce'
# )

# df['service fee'] = pd.to_numeric(
#     df['service fee'].replace('\$', '', regex=True),
#     errors='coerce'
# )

df['price'] = df['price'].replace('\$', '', regex=True).astype(float)
df['service fee'] = df['service fee'].replace('\$', '', regex=True).astype(float)

```

```

<>:13: SyntaxWarning: invalid escape sequence '\$'
<>:14: SyntaxWarning: invalid escape sequence '\$'

```

```
<>:13: SyntaxWarning: invalid escape sequence '\$'
<>:14: SyntaxWarning: invalid escape sequence '\$'
/tmp/ipykernel_8753/371903112.py:13: SyntaxWarning: invalid escape sequence '\$'
  df['price'] = df['price'].replace(['\$',],'',regex=True).astype(float)
/tmp/ipykernel_8753/371903112.py:14: SyntaxWarning: invalid escape sequence '\$'
  df['service fee'] = df['service fee'].replace(['\$',],'',regex=True).astype(float)
```

```
df.head()
```

	id	NAME	host id	host_identity_verified	host name	neighbourhood group	neighbourhood	lat	long	country	...	Construction year	price	service fee
0	1001254	Clean & quiet apt home by the park	80014485718	unconfirmed	Madaline	Brooklyn	Kensington	40.64749	-73.97237	United States	...	2020.0	966.0	193.0
1	1002102	Skylit Midtown Castle	52335172823	verified	Jenna	Manhattan	Midtown	40.75362	-73.98377	United States	...	2007.0	142.0	28.0
2	1002403	THE VILLAGE OF HARLEM....NEW YORK !	78829239556	NaN	Elise	Manhattan	Harlem	40.80902	-73.94190	United States	...	2005.0	620.0	124.0
4	1003689	Entire Apt: Spacious Studio/Loft by central park	92037596077	verified	Lyndon	Manhattan	East Harlem	40.79851	-73.94399	United States	...	2009.0	204.0	41.0
5	1004098	Large Cozy 1 BR Apartment In Midtown East	45498551794	verified	Michelle	Manhattan	Murray Hill	40.74767	-73.97500	United States	...	2013.0	577.0	115.0

5 rows × 24 columns

Remove Duplicates

```
df.drop_duplicates(inplace=True)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 101410 entries, 0 to 102057
Data columns (total 24 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   id                                     101410 non-null  int64
1   NAME                                  101410 non-null  object
2   host id                               101410 non-null  int64
3   host_identity_verified                101134 non-null  object
4   host name                             101410 non-null  object
5   neighbourhood group                   101384 non-null  object
6   neighbourhood                         101394 non-null  object
7   lat                                   101402 non-null  float64
8   long                                  101402 non-null  float64
9   country                               100884 non-null  object
10  country code                          101288 non-null  object
```

```

11 instant_bookable      101314 non-null object
12 cancellation_policy    101340 non-null object
13 room type              101410 non-null object
14 Construction year      101210 non-null float64
15 price                  101171 non-null float64
16 service fee            101142 non-null float64
17 minimum nights         101016 non-null float64
18 number of reviews      101228 non-null float64
19 last review            101410 non-null datetime64[ns]
20 reviews per month      101410 non-null float64
21 review rate number     101103 non-null float64
22 calculated host listings count 101092 non-null float64
23 availability 365       100990 non-null float64
dtypes: datetime64[ns](1), float64(11), int64(2), object(10)
memory usage: 19.3+ MB

```

Descriptive statics

```
df.describe()
```

	id	host id	lat	long	Construction year	price	service fee	minimum nights	number of reviews	last review	reviews per month	review rate
count	1.014100e+05	1.014100e+05	101402.000000	101402.000000	101210.000000	101171.000000	101142.000000	101016.000000	101228.000000	101410	101410.000000	101103
mean	2.920959e+07	4.926155e+10	40.728082	-73.949663	2012.486908	625.381008	125.043998	8.113744	27.511854	2018-05-15 21:26:08.721033728	1.163207	3.111854
min	1.001254e+06	1.236005e+08	40.499790	-74.249840	2003.000000	50.000000	10.000000	-1223.000000	0.000000	2012-07-11 00:00:00	0.000000	1.000000
25%	1.507574e+07	2.459183e+10	40.688730	-73.982570	2007.000000	340.000000	68.000000	2.000000	1.000000	2017-07-30 00:00:00	0.090000	2.000000
50%	2.922911e+07	4.912069e+10	40.722300	-73.954440	2012.000000	625.000000	125.000000	3.000000	7.000000	2019-05-23 00:00:00	0.480000	3.000000
75%	4.328308e+07	7.399747e+10	40.762750	-73.932340	2017.000000	913.000000	183.000000	5.000000	31.000000	2019-07-01 00:00:00	1.710000	4.000000
max	5.736742e+07	9.876313e+10	40.916970	-73.705220	2022.000000	1200.000000	240.000000	5645.000000	1024.000000	2058-06-16 00:00:00	90.000000	5.000000
std	1.626820e+07	2.853703e+10	0.055850	0.049474	5.765130	331.609111	66.313374	30.378014	49.549258	NaN	1.683708	1.000000

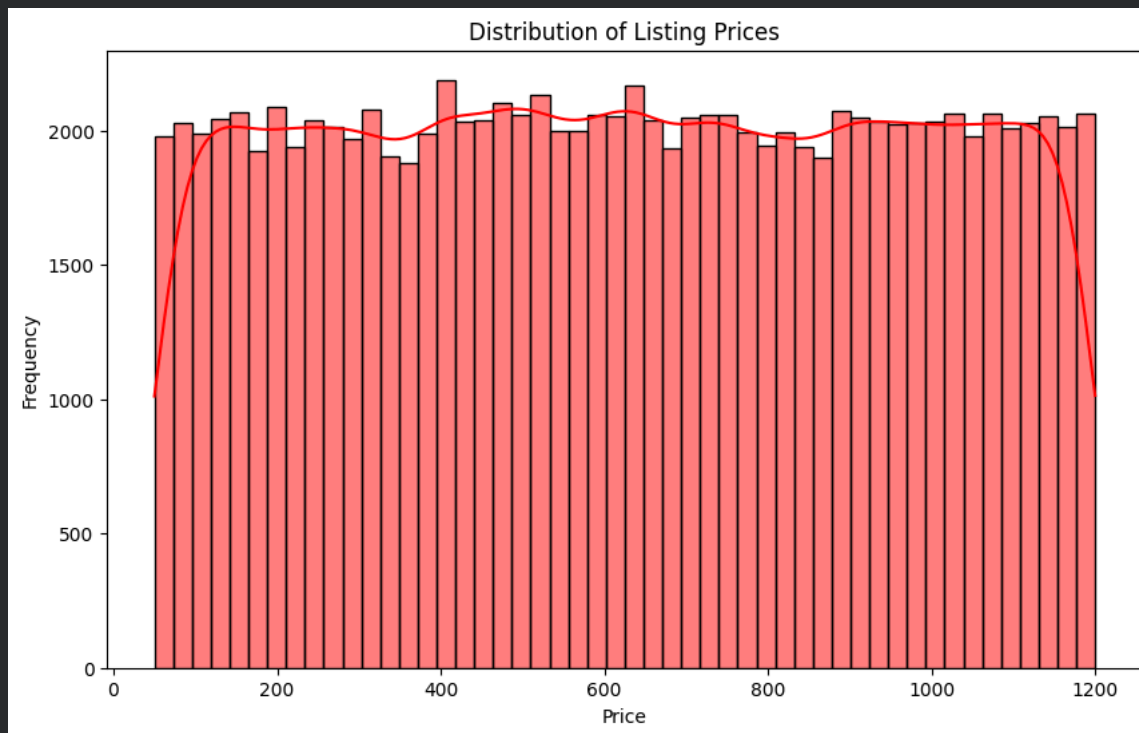
Visualization

1. What is the distribution of listing prices?

```

plt.figure(figsize=(10,6))
sns.histplot(df['price'],bins=50,kde=True,color='red')
plt.title('Distribution of Listing Prices')
plt.xlabel('Price')
plt.ylabel('Frequency')
plt.show()

```



The histogram shows that Airbnb listing prices are spread across many different price ranges. There is no one price range where most listings are grouped. The KDE line also shows this spread clearly, meaning the dataset has listings with different price levels.

2. How are different room types distributed?

```
df['room type']
```

```
room type
0      Private room
1      Entire home/apt
2      Private room
4      Entire home/apt
5      Entire home/apt
...      ...
102053  Private room
102054  Private room
102055  Entire home/apt
102056  Private room
102057  Entire home/apt
101410 rows x 1 columns
```

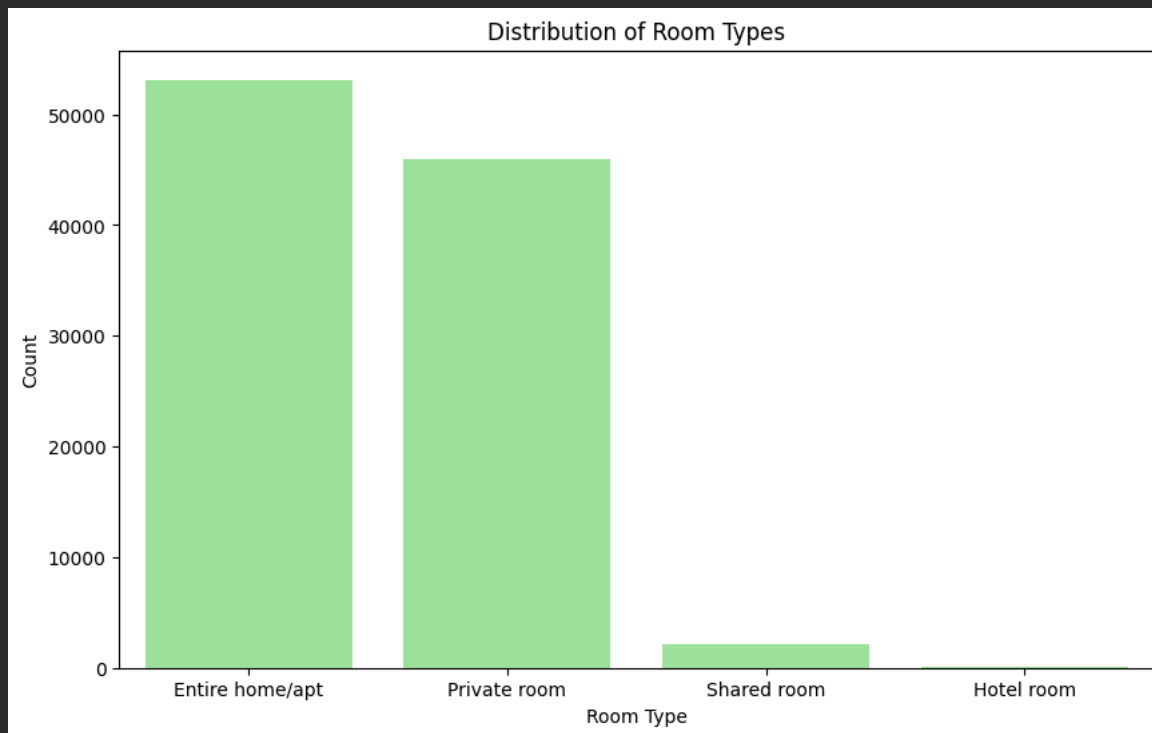
dtype: object

```
df['room type'].value_counts()
```

```
count
room type
Entire home/apt  53132
Private room    45967
Shared room     2196
Hotel room      115
```

dtype: int64

```
plt.figure(figsize=(10,6))
sns.countplot(x='room type',data=df,color="lightgreen", order=df['room type'].value_counts().index)
plt.title('Distribution of Room Types')
plt.xlabel('Room Type')
plt.ylabel('Count')
plt.show()
```



3. How are listings distributed across different neighborhoods?

```
df['neighbourhood group']
```

```
neighbourhood group
```

0	Brooklyn
1	Manhattan
2	Manhattan
4	Manhattan
5	Manhattan
...	...
102053	Brooklyn
102054	Brooklyn
102055	Brooklyn
102056	Manhattan
102057	Manhattan

101410 rows × 1 columns

dtype: object

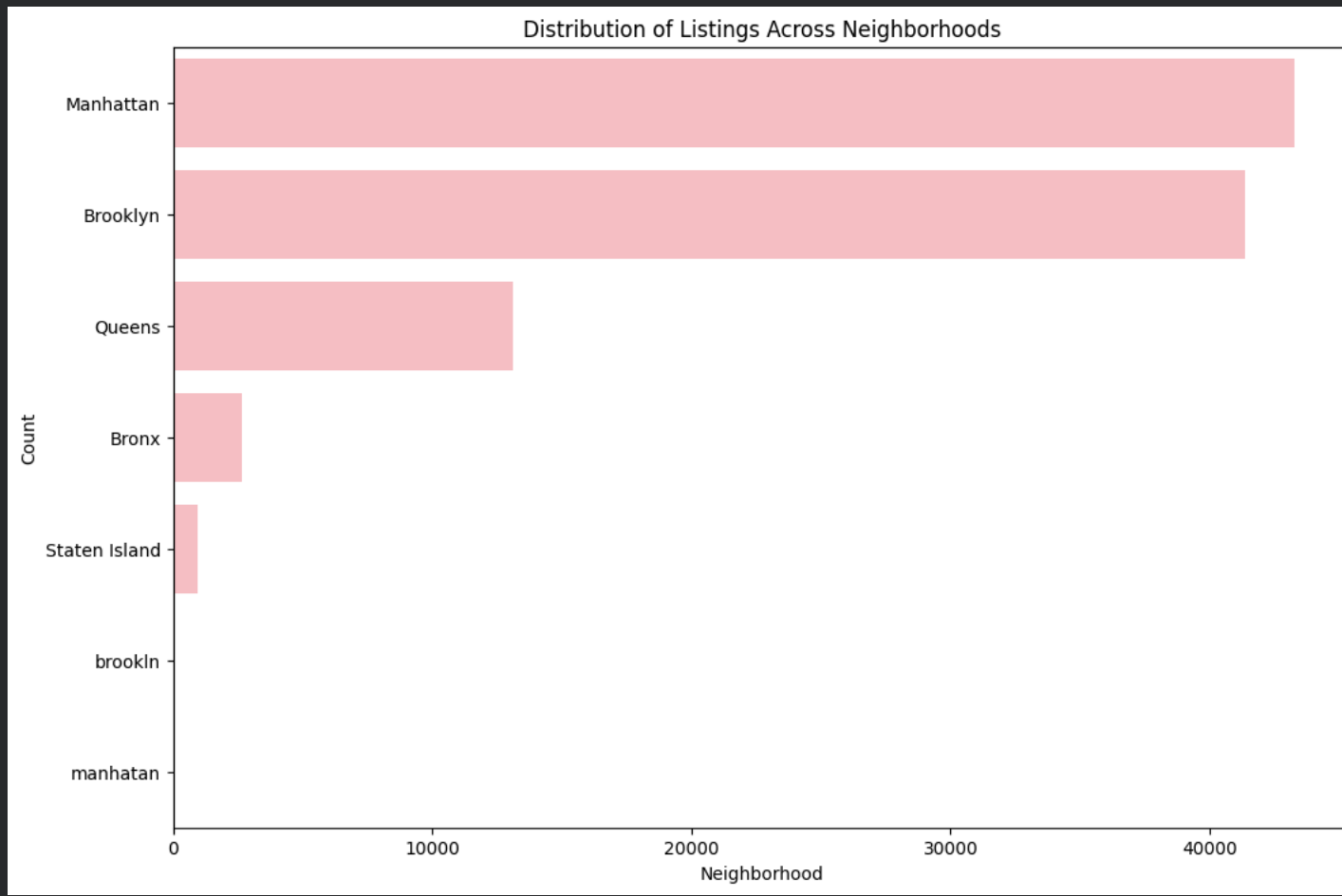
```
df['neighbourhood group'].value_counts()
```

```
count
```

neighbourhood group	count
Manhattan	43278
Brooklyn	41363
Queens	13120
Bronx	2678
Staten Island	943
brookln	1
manhatan	1

dtype: int64

```
plt.figure(figsize=(12,8))
sns.countplot(y='neighbourhood group',data=df,color="lightpink" , order=df['neighbourhood group'].value_counts().index)
plt.title('Distribution of Listings Across Neighborhoods')
plt.xlabel('Neighborhood')
plt.ylabel('Count')
plt.show()
```



4. What is the relationship between price and room type?

```
df['price'].describe()
```

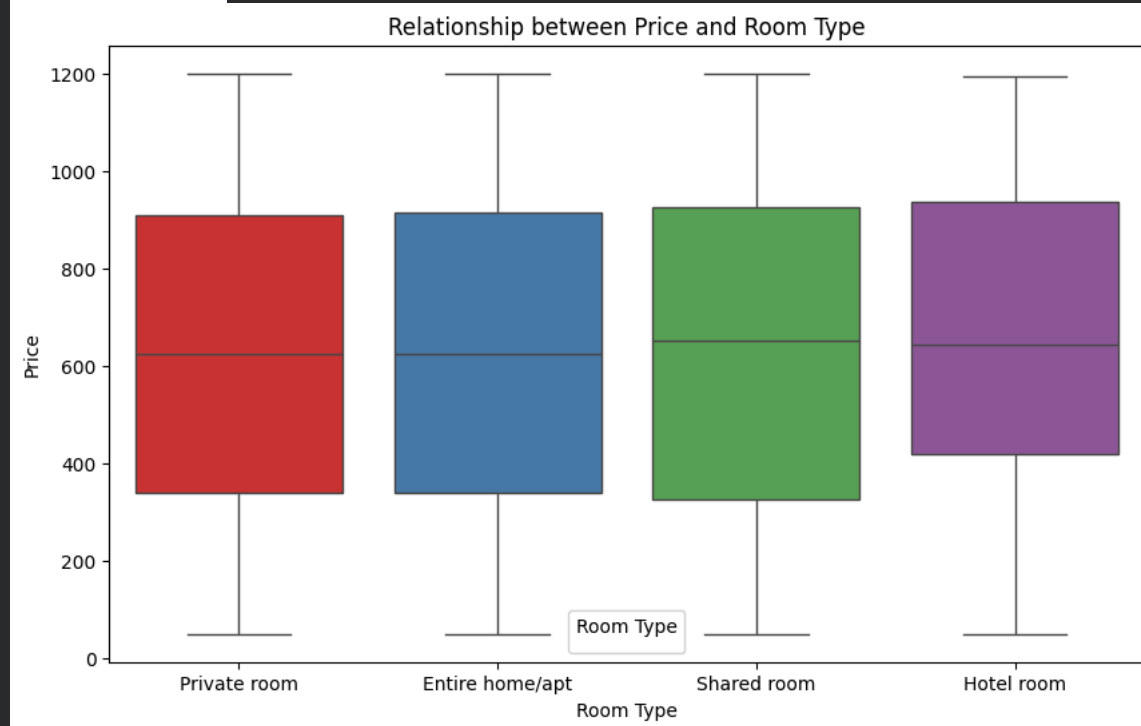
```

price
count 101171 000000

plt.figure(figsize=(10,6))
sns.boxplot(x='room type',y='price',hue='room type',data=df,palette='Set1')
plt.title('Relationship between Price and Room Type')
plt.xlabel('Room Type')
plt.ylabel('Price')
plt.legend(title='Room Type')
plt.show()

30% 025.000000
75% 018.000000
100% 025.000000
/tmp/ipykernel_9753/1948448832.py:6: UserWarning: No artists with labels found to put in legend. Note that artists whose label start with an underscore are ignored.
  plt.legend(title='Room Type')

```



5. How has the number of reviews changed over time?

```

df['last review'] = pd.to_datetime(df['last review'])
reviews_over_time = df.groupby(df['last review'].dt.to_period('M')).size()

```

```
# reviews_over_time.index.name = None
```

```
# reviews over time columns = ['Month' 'Number of Reviews']
```